7.5 Storing objects programmatically

S3 is accessible using an API via HTTPS. This enables you to integrate S3 into your applications by making requests to the API programmatically. Doing so allows your applications to benefit from a scalable and highly available data store. AWS offers free SDKs for common programming languages like C++, Go, Java, JavaScript, .NET, PHP, Python, and Ruby. You can execute the following operations using an SDK directly from your application:

- Listing buckets and their objects
- Creating, removing, updating, and deleting (CRUD) objects and buckets
- Managing access to objects

Here are examples of how you can integrate S3 into your application:

- *Allow a user to upload a profile picture.* Store the image in S3, and make it publicly accessible. Integrate the image into your website via HTTPS.
- *Generate monthly reports (such as PDFs), and make them accessible to users.* Create the documents and upload them to S3. If users want to download documents, fetch them from S3.
- *Share data between applications.* You can access documents from different applications. For example, application A can write an object with the latest information about sales, and application B can download the document and analyze the data.

Integrating S3 into an application is one way to implement the concept of a *stateless server*. We'll show you how to integrate S3 into your application by diving into a simple web application called Simple S3 Gallery next. This web application is built on top of Node.js and uses the AWS SDK for JavaScript and Node.js. You can easily transfer what you learn from this example to SDKs for other programming languages; the concepts are the same.

Installing and getting started with Node.js